

SRIKANTH NAYAK, Ph.D.

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RESEARCH INTERESTS

Separation science, colloidal and interfacial phenomena, functional nanomaterials

HIGHLIGHTS

- **Interfaces in separation systems:** Investigated ion adsorption at charged 2D interfaces (Langmuir monolayers) as model systems for selective separations. Discovered the spontaneous formation of inverted bilayers at air/aqueous interfaces and the essential role of interfacial ion-hydration in solvent extraction.
- **Mesoscale structure of complex fluids:** Investigated the mesoscale structure of complex fluids containing tertiary amines using SAXS and spectroscopy. Developed analytical models to better understand X-ray scattering from aggregated complex fluids containing ammonium-based organic cations.
- **Nanomaterial synthesis:** Pioneered hierarchical self-assembly of nanoparticle superlattices using interpolymer complexation and demonstrated templated assembly of nanoparticles by leveraging proteins and polymers. Established scalable methods for creating highly ordered nanoparticle superlattices by modulating solution conditions.
- **Advanced material characterization:** Highly proficient in synchrotron techniques including X-ray reflectivity (XRR), SAXS/WAXS, and GISAXS to map interfacial structures.
- **f-element coordination:** Established molecular mechanisms for lanthanide and actinide extraction and speciation in complex fluid environments. Conceptualized and executed separation chemistry studies of lanthanides in thiocyanate media, directly supporting research into separation of lanthanides and radionuclides; Synthesized Ce(IV) and Pu(IV) oxide nanoparticles and connected their phase behaviour in concentrated aqueous electrolytes with their surface chemistry and morphological features.
- **Publication record:** Authored 23 peer-reviewed articles, including first-author works in *Nanoscale*, *Langmuir*, *ACS Applied Materials & Interfaces*, *JPCL*, and *JPCC*.
- **Mentoring and professional service:** Mentored undergraduate students and SULI interns in analytical chemistry, experimental design, lab safety, and data analysis; Served as a reviewer for the US DOE American-Made Prize (2024) and high-impact journals including *Small*, *Langmuir*, and *Separations*.

EDUCATION

Iowa State University, Ames, IA

Ph.D., Chemical Engineering, 2018

Dissertation: Macromolecular Soft Templates for Synthesis and Hierarchical Self-Assembly of Functional Hybrid Materials

Advisor: Dr. Surya Mallapragada

Indian Institute of Technology, Kharagpur, Kharagpur, West Bengal, India

Dual Degree (B. Tech and M. Tech), Chemical Engineering, 2013

Dissertation: Adaptive Mesh Refinement in Modelling In Situ Combustion for Enhanced Oil Recovery

Advisor: Dr. Somenath Ganguly

PROFESSIONAL EXPERIENCE

- 2021-2025 *Assistant Chemist*, Heavy Elements and Separation Science, Argonne National Laboratory, Lemont, IL.
- 2019-2021 *Postdoctoral Appointee*, Heavy Element Chemistry and Separation Science, Argonne National Laboratory, Lemont, IL. *Supervisor*: Dr. Ahmet Uysal
- 2013-2018 *Graduate Research Assistant*, Dept. of Chemical Engineering, Iowa State University, Ames, IA. *Supervisor*: Dr. Surya Mallapragada
- 2011-2013 *Undergraduate Research Assistant*, Dept. of Chemical Engineering, IIT Kharagpur, West Bengal, India. *Supervisor*: Dr. Somenath Ganguly
- 2011 *Summer Intern*, Rashtriya Chemicals and Fertilizers, Chembur, India

MENTORING AND TEACHING EXPERIENCE

- Guest lecturer for MDI 471 – Materials for Regenerative Economy, Fall 2025, U of Buffalo
- Supervised three undergraduate interns supported by the SULI program of U.S DoE in conducting analytical chemistry research. Summer, 2022 and Summer, 2023.
- Teaching Assistant, Dept. of Chemical Engineering, Iowa State University, Fall 2014, Spring 2015
Courses: Computational Methods for Chemical Engineers, Chemical Engineering Laboratory
- Teaching Assistant, Dept. of Chemical Engineering, IIT Kharagpur, Fall 2012
Course: Process Equipment Design
- Research Mentor, Mallapragada Lab, Dept. of Chemical Engineering, Iowa State University, Spring 2015 – Fall 2018
Mentored six undergraduate students in various projects in the lab, instructed them on safe conduct of experiments and critical literature review.

TECHNICAL SKILLS

- Synchrotron X-ray scattering and fluorescence techniques (SAXS/WAXS, EXAFS, GI-SAXS, GI-diffraction, and X-ray reflectometry)
- Analytical tools – ICP-MS, NMR, FTIR, Raman, Fluorescence, UV-vis, Karl-Fischer titration, Autotitrators, and liquid scintillation counting
- Materials characterization – TGA, DSC, SEM, TEM, AFM, XPS, DLS, and zeta-potential
- Surface science - Langmuir blotdgetry, drop shape analysis, PZC titrations, and vibrational SFG spectroscopy
- Polymer synthesis and characterization (ATRP, SI-ATRP, GPC, and rheometry)
- Programming languages: Python (chemometrics), MATLAB, OpenSCAD, and R | [GitHub](#)

PEER REVIEWED PUBLICATIONS

1. **Nayak, S.**; and Uysal, A. "Trivalent Rare Earth Adsorption at Phosphonic Acid Monolayers", *ChemPhysChem*, **2025**, 26, e2500429, [\[doi\]](#)
2. **Nayak, S.**, "Spectroscopic investigation of cation effects in U(VI)-NO₃⁻ complexation in aqueous solutions", *RSC Advances*, **2025**, 15, 39554-39565 [\[doi\]](#)
3. **Nayak, S.**, Hydration-Complexation Balance in Eu(III)/Er(III) Extraction from Thiocyanate Media by Aliquat-336, Under revision
4. Ito, T; Frenk, S. E.; Rai, N.; Kameta, N; Seifert, S.; Lin X.; and **Nayak, S.** "Self-Assembled Bolaamphiphile-Based Organic Nanotubes as Efficient Cu(II) Ion Adsorbents", *Langmuir*, **2024**, 40, 27377–27385 [\[doi\]](#)
5. **Nayak, S.**; Servis, M. J.; Combs, D.; Szeliga, K.; and Seifert, S., "Speciation and organic phase structure in nitric acid extraction with trioctylamine", *The Journal of Physical Chemistry B*, **2024**, 128, 3236–3248 [\[doi\]](#)

6. Wang, X.; **Nayak, S.**; Wilson, R. E.; Soderholm, L.; and Servis, M. J. "Solvent effects on extractant conformational energetics in liquid–liquid extraction: a simulation study of molecular solvents and ionic liquids", *Physical Chemistry Chemical Physics*, **2024**, 26, 2877-2886 [[doi](#)]
7. Nayak, B. P.; Kim, H. J.; **Nayak, S.**; Wang, W.; Bu, W.; Mallapragada, S. K.; and Vaknin, D., Assembling PNIPAM-capped Gold Nanoparticles in Aqueous solutions, *ACS Macro Letters*, **2023**, 12, 1659-1664. [[doi](#)]
8. **Nayak, S.**; Kumal, R. R.; Lee, S. E.; Uysal, A.; Elucidating Trivalent Ion Adsorption at Floating Carboxylic Acid Monolayers: Charge Reversal or Water Reorganization?, *The Journal of Physical Chemistry Letters*, **2023**, 14, 3685–3690. [[doi](#)]
9. Bonnett, B. L.; Sheyfer, D.; Wimalasiri, P. N.; **Nayak, S.**; Lal, J.; Zhang, Q.; Seifert, S.; Stephenson, G. B.; Servis, M. J.; Critical Fluctuations in Liquid-Liquid Extraction Organic Phases Controlled by Extractant and Diluent Molecular Structure, *Physical Chemistry Chemical Physics* **2023**, 25, 16389-16403. [[doi](#)]
10. **Nayak, S.**; Kumal, R. R.; Uysal, A.; Spontaneous and Ion-Specific Formation of Inverted Bilayers at Air/Aqueous Interface, *Langmuir*, **2022**, 38, 5617–5625. [[doi](#)]
11. **Nayak, S.**; Kumal, R. R.; Liu, Z.; Qiao, B.; Clark, A.; Uysal, A.; Origins of Clustering of Metalate-Extractant Complexes in Liquid-Liquid Extraction, *ACS Applied Materials & Interfaces* **2021** 13, 24194–24206. [[doi](#)]
12. Servis, J. M.; **Nayak, S.**; Seifert, S.; The Pervasive Impact of Critical Fluctuations in Liquid-Liquid Extraction Organic Phases, *The Journal of Chemical Physics* **2021**, 155, 244506 [[doi](#)]
13. Kumal, R. R.; **Nayak, S.**; Bu, W.; Uysal, A., Chemical Potential Driven Reorganization of Anions between Stern and Diffuse Layers at the Air-Water Interface, *The Journal of Physical Chemistry C* **2022**, 126, 2, 1140–1151 [[doi](#)]
14. **Nayak, S.**; Lovering, K.; Uysal, A., Ion-Specific Clustering of Metal–Amphiphile Complexes in Rare Earth Separations. *Nanoscale* **2020**, 12, 20202-20210. [[doi](#)]
15. **Nayak, S.**; Lovering, K.; Bu, W.; Uysal, A., Anions Enhance Rare Earth Adsorption at Negatively Charged Surfaces. *The Journal of Physical Chemistry Letters* **2020**, 11, 4436-4442. [[doi](#)]
16. Lovering, K.; **Nayak, S.**; Bu, W.; Uysal, A., The Role of Specific Ion Effects in Ion Transport: The Case of Nitrate and Thiocyanate. *The Journal of Physical Chemistry C* **2020**, 124, 573-581. [[doi](#)]
17. Horst, N.; **Nayak, S.**; Wang, W. J.; Mallapragada, S.; Vaknin, D.; Travesset, A., Superlattice Assembly by Interpolymer Complexation. *Soft Matter* **2019**, 15, 9690-9699. [[doi](#)]
18. Londono-Calderon, A.; **Nayak, S.**; Mosher, C. L.; Mallapragada, S. K.; Prozorov, T., New Approach to Electron Microscopy Imaging of Gel Nanocomposites In-Situ. *Micron* **2019**, 120, 104-112. [[doi](#)]
19. **Nayak, S.**; Fieg, M.; Wang, W. J.; Bu, W.; Mallapragada, S.; Vaknin, D., Effect of (Poly)electrolytes on the Interfacial Assembly of Poly(ethylene glycol)-Functionalized Gold Nanoparticles. *Langmuir* **2019**, 35, 2251-2260. [[doi](#)]
20. **Nayak, S.**; Horst, N.; Zhang, H. H.; Wang, W. J.; Mallapragada, S.; Travesset, A.; Vaknin, D., Interpolymer Complexation as a Strategy for Nanoparticle Assembly and Crystallization. *The Journal of Physical Chemistry C* **2019**, 123, 836-840. [[doi](#)]
21. **Nayak, S.**; Horst, N.; Zhang, H. H.; Wang, W. J.; Mallapragada, S.; Travesset, A.; Vaknin, D., Ordered Networks of Gold Nanoparticles Crosslinked by Dithiol-Oligomers. *Particle & Particle Systems Characterization* **2018**, 35, 7. [[doi](#)]
22. Zhang, H. H.; **Nayak, S.**; Wang, W. J.; Mallapragada, S.; Vaknin, D., Interfacial Self-Assembly of Polyelectrolyte-Capped Gold Nanoparticles. *Langmuir* **2017**, 33, 12227-12234. [[doi](#)]
23. **Nayak, S.**; Zhang, H. H.; Liu, X. P.; Feng, S. R.; Palo, P.; Nilsen-Hamilton, M.; Akinc, M.; Mallapragada, S., Protein Patterns Template Arrays of Magnetic Nanoparticles. *RSC Advances* **2016**, 6, S7048-S7056. [[doi](#)]
24. Liu, X. P.; Zhang, H. H.; **Nayak, S.**; Parada, G.; Anderegg, J.; Feng, S. R.; Nilsen-Hamilton, M.; Akinc, M.; Mallapragada, S. K., Effect of Surface Hydrophobicity on the Function of the Immobilized Biomineralization Protein Mms6. *Industrial & Engineering Chemistry Research* **2015**, 54, 10284-10292. [[doi](#)]

HONORS & AWARDS

- Chevron Phillips Graduate Fellowship, 2013
Full funding for the first year of Ph.D.
- Advanced Photon Source User Meeting Travel Award, 2018

SERVICE

- *Reviewer* for a) Langmuir (ACS), b) Small (Wiley), c) Particle and Particle Systems Characterization (Wiley), d) Separations (MDPI), e) Colloids and Interfaces (MDPI), f) Applied Sciences (MDPI), and g) Food (MDPI)
- *Reviewer* for American-Made Electronics Scrap Recycling Advancement Prize, US DoE AMMTO, 2024
- *Review Editor* for *Frontiers in Chemistry (Physical Chemistry and Chemical Physics)*
- *Co-chair*, Postdoctoral Research and Career Symposium, Argonne National Laboratory 2020
- *Treasurer*, Chemical Engineering Graduate Students Organization, Iowa State University, 2014-2015.
- *Volunteer Instructor*, CAPS High School Computing Workshop, Argonne National Laboratory 2019
- *Project Coordinator*, Sankalp, Iowa State University, 2014-2015